

Macroeconomics III

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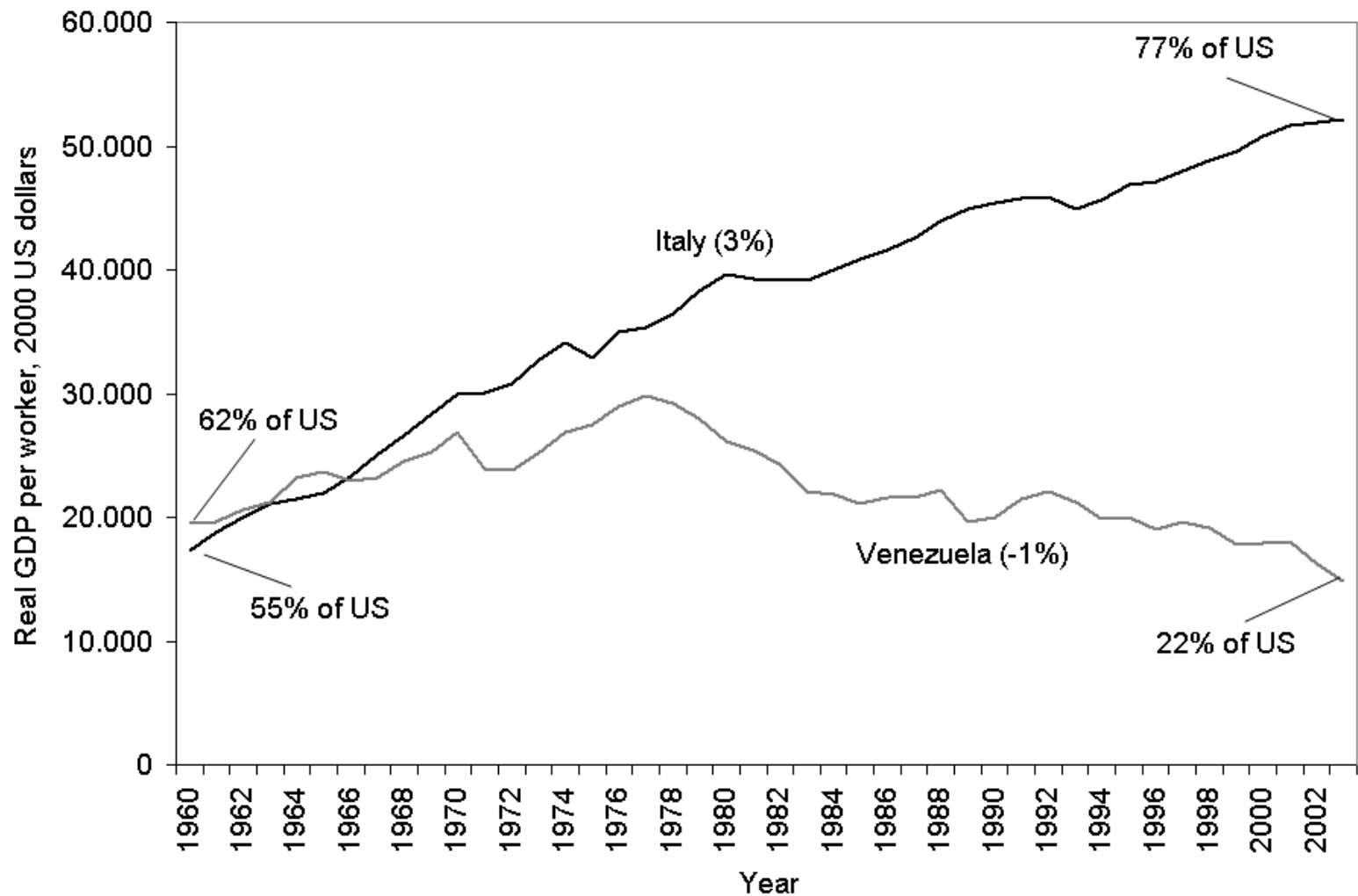
based on Chapter 2
of your Textbook

“SOME FACTS
ABOUT PROSPERITY
AND GROWTH”

INTRODUCTION

- How do we measure prosperity? In economics, often by GDP per person!
- Higher income means higher consumption or higher saving (possibly both) and saving can be used for future consumption. Consumption is always rooted in income and is a decent proxy for well-being.
- In this course we often consider **growth** of income per person because the way to reach a high **level** of income per person is through a process of high growth in income per person.
- Relatively small differences in growth rates can imply large differences in levels over long periods as evidenced by next slide.
- Conclusion: Growth is important! One focus of our classes is **What creates growth?**

The importance of growth



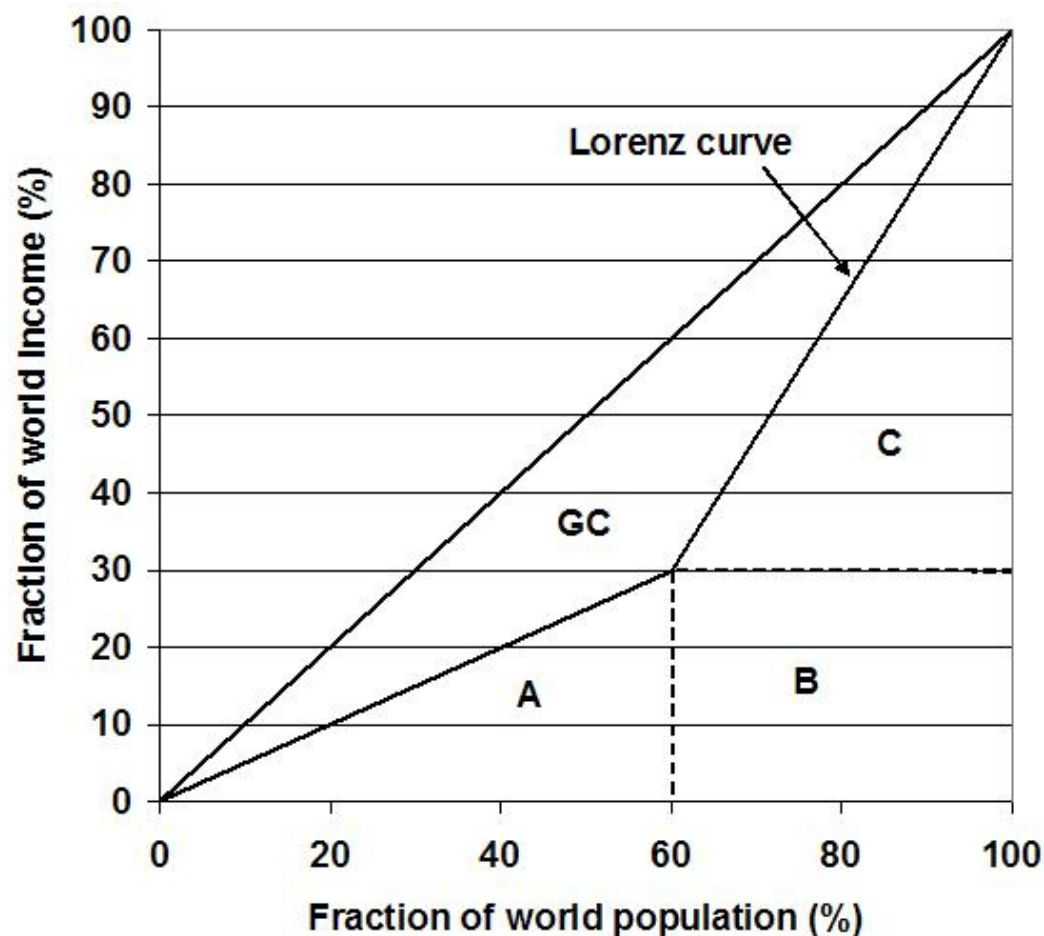
COMPARING THE GDP LEVELS OF TWO COUNTRIES

- Purchasing-power adjustment:
 - When converting, say, pesos into US dollars, we want to take into account differences in the costs of living in the two countries.
 - Hence, the relevant rate-of-conversion should reflect that consumer goods are usually much cheaper in less developed countries.
- Per worker or per capita?
 - GDP per capita = Official GDP divided by total population:
Underestimates the production of less developed countries, since they typically have a larger share of the population working outside the official market economy.
 - GDP per worker = Official GDP divided by the labour force (population times participation rate): corrects for differences in participation rates, hence it is more of a productivity measure.

THE LORENZ CURVE

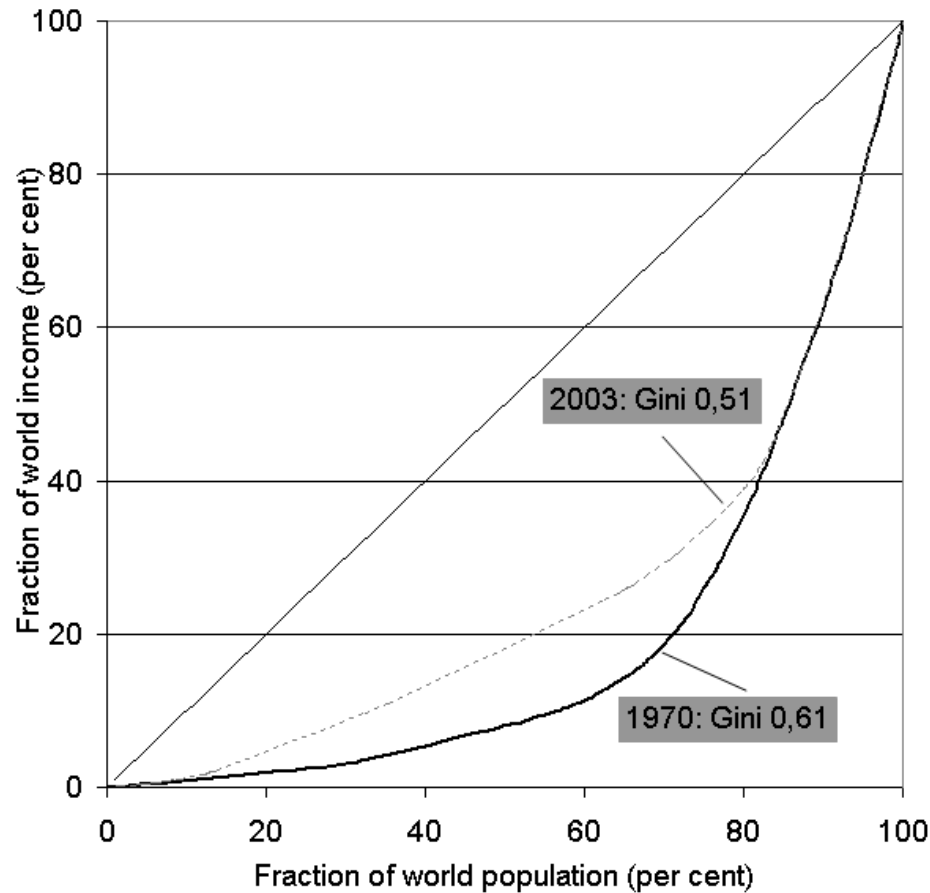
- ... is a comprehensive way to measure the degree of inequality of a personal income distribution, for a single country or for the world.
- Stylized example: Suppose there are two countries in the world, one rich and one poor and that 60% of the world population lives in the poor country earning 30% of world income, while 40% lives in the rich country earning 70% of world income.
- The Lorenz curve is defined as the curve through a collection of points (x, y) , where x is the share of people with incomes *below* a certain level and y is the income share of these people.
- The curve always starts at $(0,0)$. In our example, the next point is $(60,30)$, since the 60% poorest people earn 30% of world income. Finally, the curve always ends in $(100,100)$. Curve on next slide.

- Imagine that the income distribution were characterized by perfect equality.
- Then the 60% poorest people would earn 60% of total income.
- The Lorenz curve would be identical to the diagonal from the lower left to the upper right corner indicated in the figure.
- We can use this to define a measure of inequality called...



THE GINI COEFFICIENT

- The area between the Lorenz curve and the diagonal divided by the full area below the diagonal. In the figure that is $GC/(GC+A+B+C)$. A measure between zero and one.
- If the Gini coefficient is zero, there is no inequality.
- The larger the Gini coefficient becomes, the more inequality there is.
- Using GDP per worker and population data for 137 countries and artificially associating with each person in a country the GDP per worker of the country, we can draw Lorenz curves for the world in 1960 and 2003, respectively:



- **Stylized fact 1:** Some countries are rich and some are poor; the differences are enormous and it has pretty much stayed like that in relative terms over the last 40 years. However, there is some tendency towards a more equal world income distribution, though not much at the very bottom.

Prosperity and poverty, growth and decline

Table 2.2: The world's prosperity 'Top 15' and 'Bottom 15', 1965 and 2003

	Real GDP per worker relative to USA		Average annual growth rate 1965-2003		Real GDP per worker relative to USA		Average annual growth rate 1965-2003
	Per cent 1965	Per cent 2003			Per cent 1965	Per cent 2003	
Poorest in 1965			Per cent	Poorest in 2003			Per cent
Ghana	1.9	4.2	3.8	Guinea-Bissau	3.4	1.9	0.1
China	2.4	12.2	6.1	Burundi	3.1	2.1	0.6
Malawi	2.5	2.4	1.4	Ethiopia	2.5	2.3	1.3
Ethiopia	2.5	2.3	1.3	Madagascar	6.5	2.3	-1.1
Tanzania	2.8	2.6	1.4	Malawi	2.5	2.4	1.4
Burkina Faso	2.9	3.4	2.0	Tanzania	2.8	2.6	1.4
Burundi	3.1	2.1	0.6	Gambia	4.3	2.7	0.3
Mali	3.1	3.6	2.0	Niger	6.8	2.7	-0.9
Guinea-Bissau	3.4	1.9	0.1	Togo	6.2	2.7	-0.6
Nepal	4.1	4.5	1.9	Chad	6.3	2.9	-0.5
Rwanda	4.2	3.5	1.1	Zambia	6.9	3.3	-0.4
Lesotho	4.2	7.0	2.9	Burkina Faso	2.9	3.4	2.0
Mozambique	4.3	4.1	1.5	Uganda	4.7	3.4	0.7
Gambia	4.3	2.7	0.3	Kenya	6.8	3.4	-0.2
Uganda	4.7	3.4	0.7	Rwanda	4.2	3.5	1.1
Richest in 1965	1965	2003		Richest in 2003	1965	2003	
Switzerland	103.0	79.9	0.9	United States	100.0	100.0	1.6
United States	100.0	100.0	1.6	Ireland	44.0	97.1	3.7
New Zealand	99.1	65.3	0.5	Norway	76.4	96.8	2.2
Netherlands	91.5	83.7	1.4	Belgium	71.4	90.7	2.2
Canada	87.5	76.3	1.2	Austria	60.2	88.1	2.6
Denmark	81.9	75.9	1.4	France	67.0	83.9	2.2
Sweden	81.2	70.9	1.2	Netherlands	91.5	83.7	1.4
Australia	78.7	80.5	1.7	Australia	78.7	80.5	1.7
Norway	76.4	96.8	2.2	Singapore	35.7	80.0	3.8
Belgium	71.4	90.7	2.2	Switzerland	103.0	79.9	0.9
Iceland	71.2	68.1	1.5	Italy	59.1	76.8	2.3
United Kingdom	67.6	76.5	1.9	United Kingdom	67.6	76.5	1.9
France	67.0	83.9	2.2	Canada	87.5	76.3	1.2
Israel	66.6	69.9	1.7	Denmark	81.9	75.9	1.4
Venezuela	63.5	21.9	-1.2	Hong Kong	37.1	75.2	3.5

Note: Countries are chosen from a sample of 97 countries, for which data on real GDP per worker in 1965 and 2003 were available. A lower limit of 1 mio. citizens was used.

Prosperity and poverty, growth and decline

Table 2.3: World growth 'bottom 20' and 'top 20', 1965–2003

	Average annual growth rate of GDP per worker 1965–2003	Real GDP per worker relative to USA, per cent	
	Per cent	1965	2003
20 slowest growing			
Nicaragua	-1.9	48.9	12.7
Jordan	-1.3	50.8	16.8
Venezuela	-1.2	63.5	21.9
Madagascar	-1.1	6.5	2.3
Niger	-0.9	6.8	2.7
Togo	-0.6	6.2	2.7
Chad	-0.5	6.3	2.9
Zambia	-0.4	6.9	3.3
Peru	-0.3	33.9	16.3
Jamaica	-0.3	27.3	13.3
Kenya	-0.2	6.8	3.4
Senegal	-0.2	9.1	4.6
Bolivia	0.0	19.2	10.7
Guinea-Bissau	0.1	3.4	1.9
El Salvador	0.1	28.8	16.1
20 fastest growing			
China	6.1	2.4	12.2
Korea, Republic of	5.2	13.4	49.8
Thailand	4.2	6.9	18.0
Malaysia	4.0	16.9	41.2
Ghana	3.8	1.9	4.2
Singapore	3.8	35.7	80.0
Ireland	3.7	44.0	97.1
Romania	3.6	8.7	18.5
Hong Kong	3.5	37.1	75.2
Japan	3.2	37.0	66.4
Sri Lanka	3.1	7.7	13.7
Indonesia	3.1	7.1	12.2
India	2.9	6.0	9.9
Lesotho	2.9	4.2	7.0
Austria	2.6	60.2	88.1

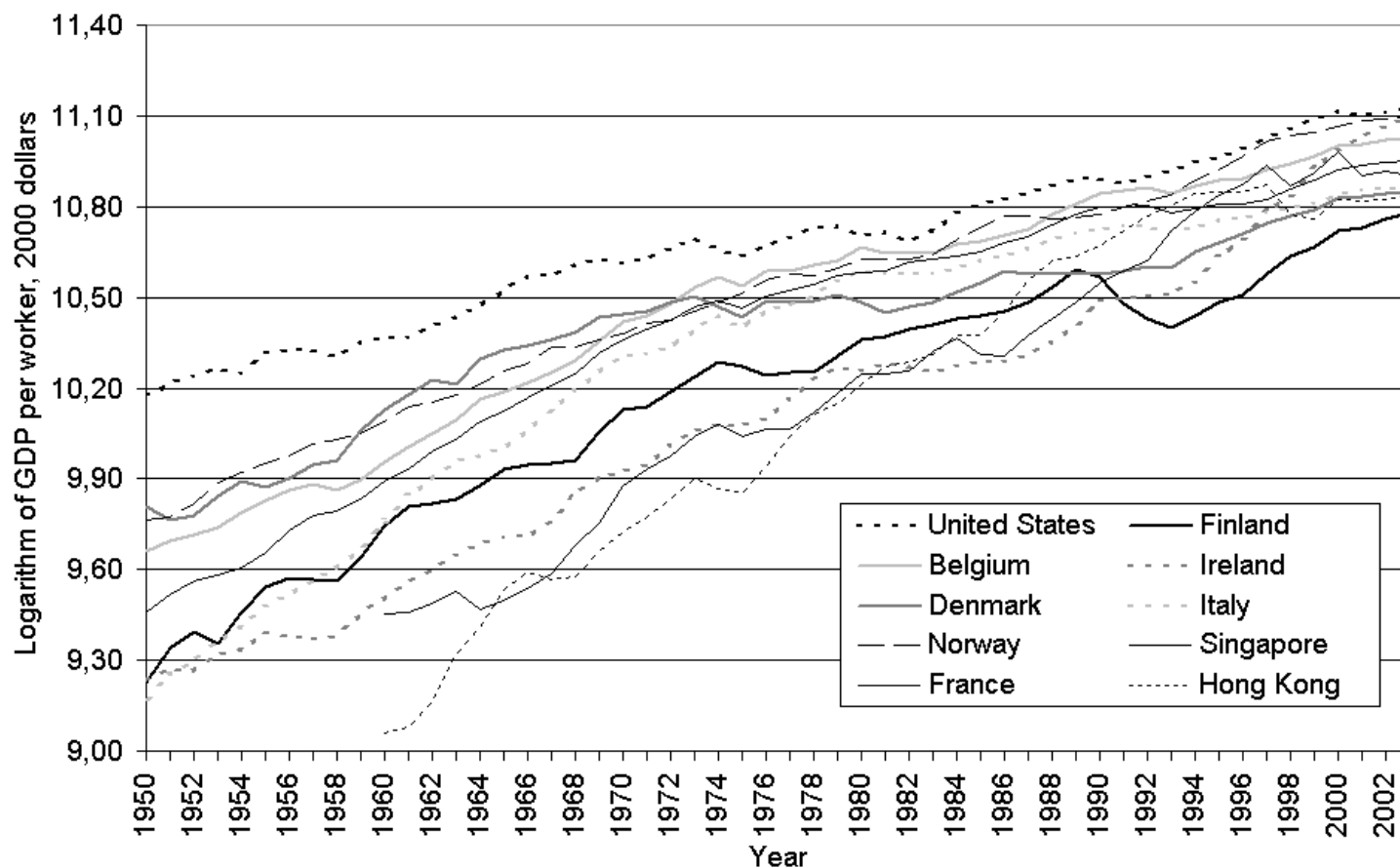
Note: Sample of 97 countries as for Table 2.2. Botswana is missing for lack of 1965 to 1970 data, while Taiwan is missing for lack of 1998–2003 data. They would otherwise have been in the top of the top 20.

- In the 'top 20' of the fastest growing countries we find the 'growth miracles', e.g., tiger economies such as Korea, Singapore, Hong Kong, China and India. In the bottom we find the 'growth disasters', many of which have had negative average growth rates over the period considered.

- **Stylized fact 2:** Growth rates vary substantially between countries, and by the process of growing or declining fast, a country can move from being relatively poor to being relatively rich, or from being relatively rich to being relatively poor.
- **Stylized fact 3:** Growth can break in a country, turning from a high rate to a low one or vice versa.
- In the 1960s the East-Asian “growth miracles” experienced growth breaks, moving from low to high growth rates. Hence, it **is** possible for poor countries to escape from poverty!

CONVERGENCE

Hypothesis of absolute convergence: In the long run, GDP per worker (or per capita) converges to one and the same growth path in all countries, so that all countries converge on the same level of income per worker.

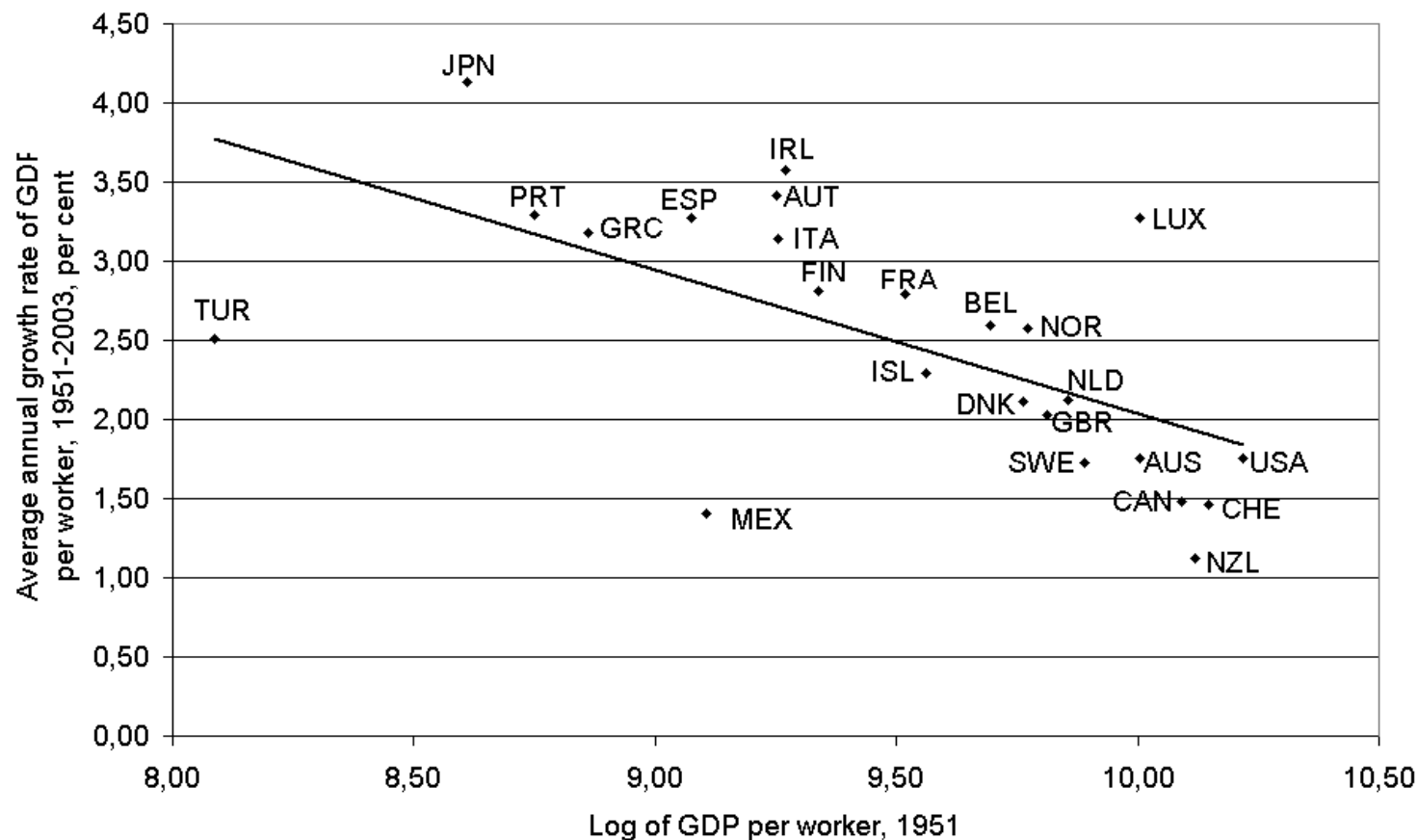


- If two countries are to end up with the same level of GDP per worker and one of them starts at a level much lower than the other, then that country must grow relatively quickly compared to the other.
- That is, if we estimate across countries i a regression line:

$$\frac{\ln y_T^i - \ln y_0^i}{T} = \beta_0 - \beta_1 \ln y_0^i$$

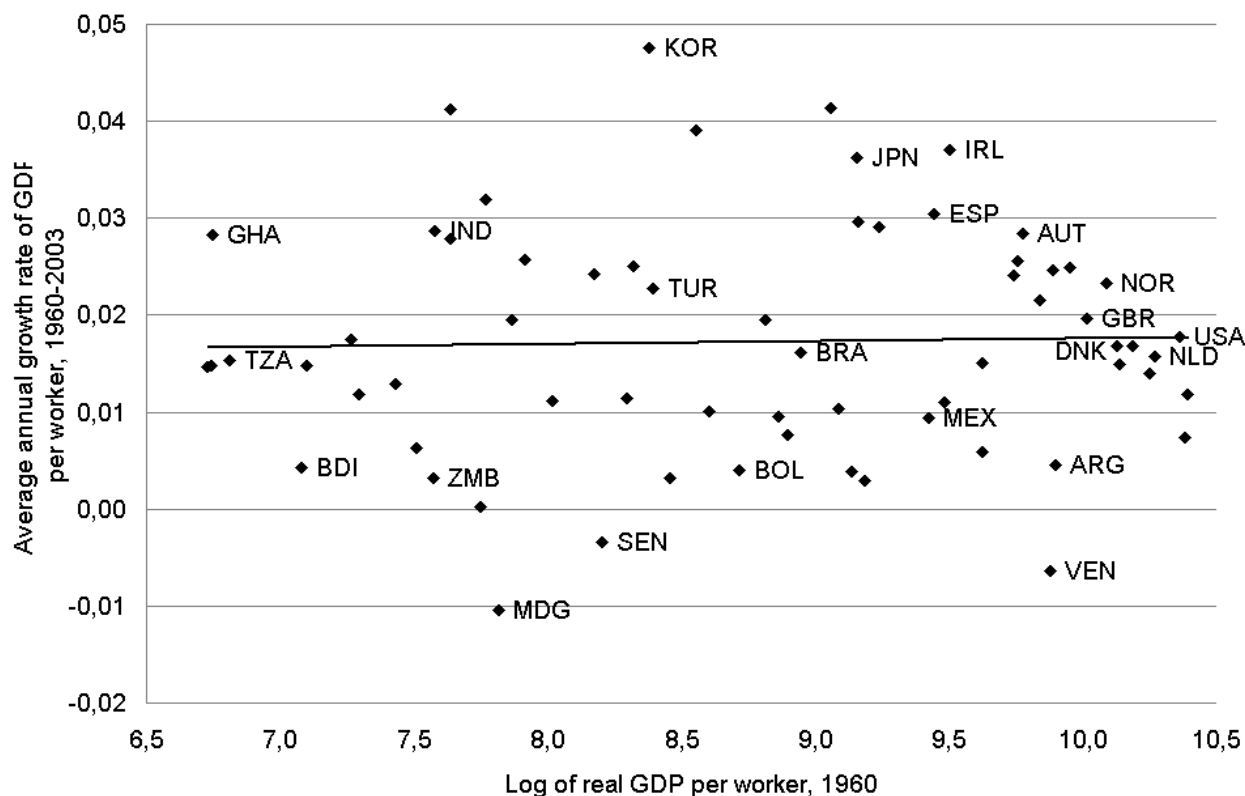
where $(\ln y_T^i - \ln y_0^i) / T$ is the average growth rate of GDP per worker, and $\ln y_0^i$ is the initial level of GDP per worker, we expect the estimate of β_1 to be significantly positive - note the minus in front of β_1 .

- For a sample of 24 rich OECD countries we find:



- There is a tight negative correlation and the estimate of β_1 is indeed significantly positive.
- However, the countries included in the sample were all relatively rich in 2003. Therefore there is a 'sample selection bias'.

- For a more representative sample of 65 countries we find:



- The negative relationship has disappeared, and β_1 is insignificant. We have to reject the hypothesis of absolute convergence.

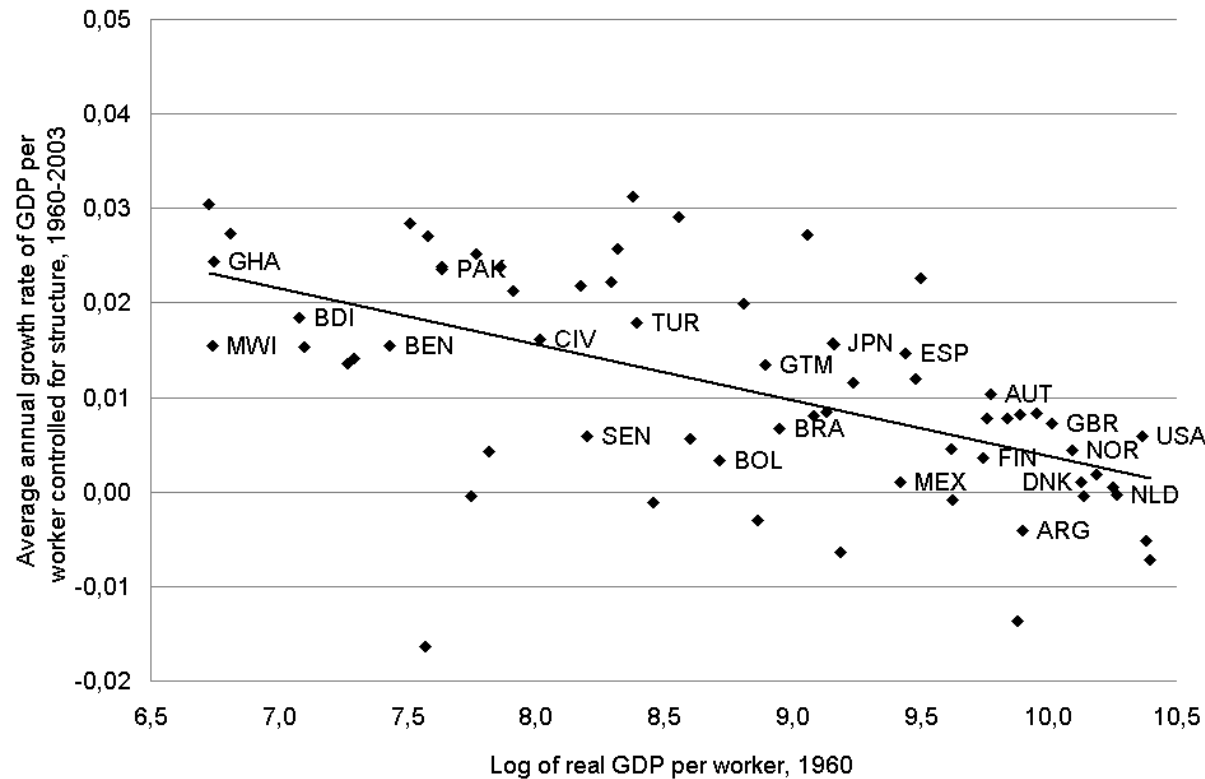
Hypothesis of conditional convergence: A country's income per worker converges to a **country-specific** long-run growth path, which depends on the basic structural characteristics of the country. The further below its own long-run growth path a country starts, the faster it will grow. Income per worker therefore converges to the same level across countries **conditional** on the countries being structurally alike.

- The "correct" regression line could be an equation like:

$$\frac{\ln y_T^i - \ln y_0^i}{T} = \beta_0 - \beta_1 \ln y_0^i + \beta_2 \left[\ln s^i - \ln (n^i + 0.075) \right]$$

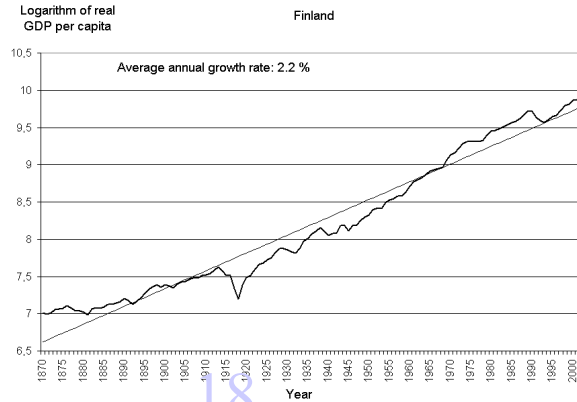
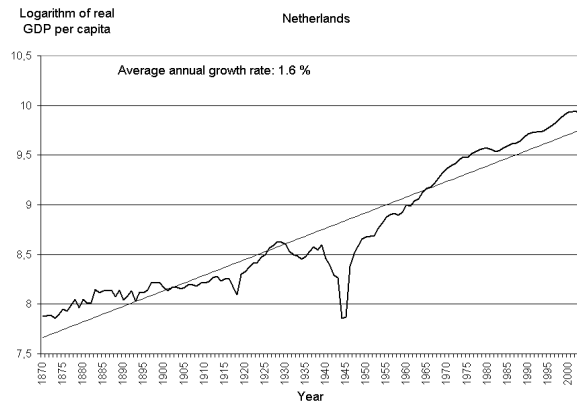
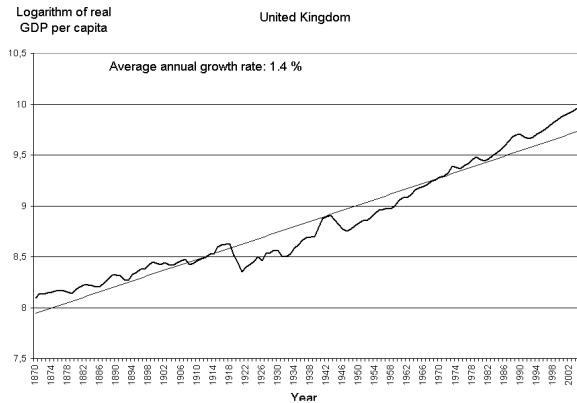
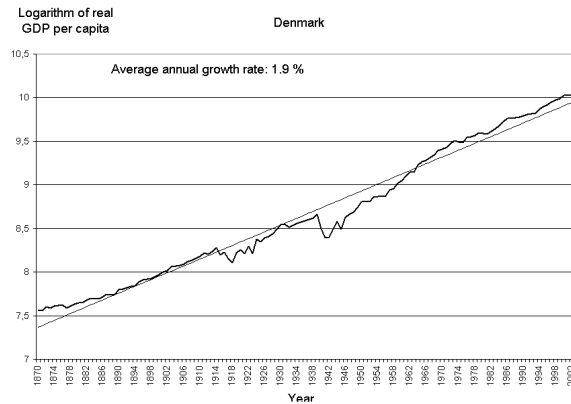
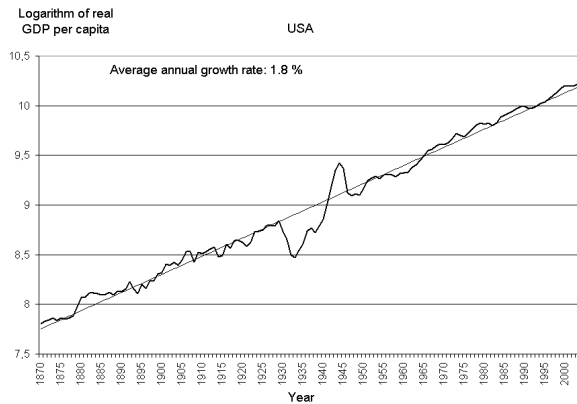
where the last term on the right hand side captures structural influence, where s^i is the average investment rate of country i , and n^i is the average growth rate of the labour force. (The theory behind this formulation is introduced in later chapters.)

- For the same representative sample of 65 countries we get:



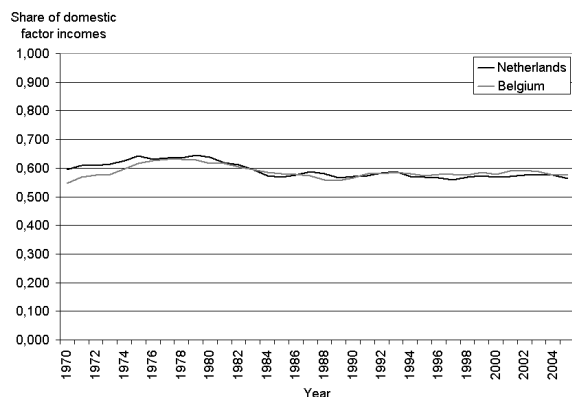
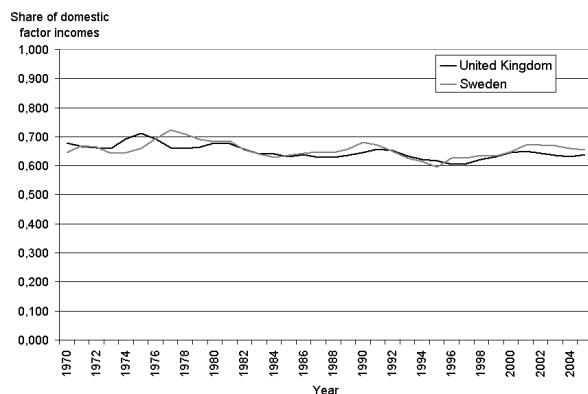
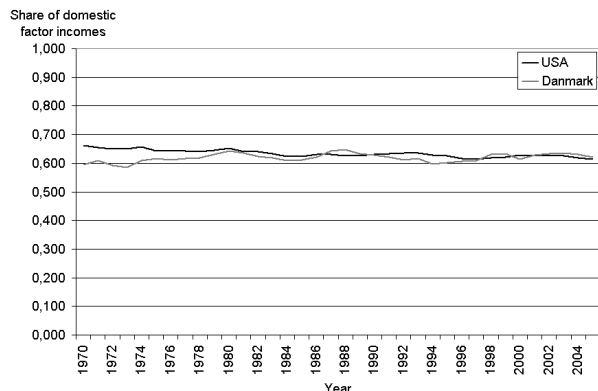
- **Stylized fact 4: Convergence:** If one controls appropriately for structural differences between the countries of the world, a lower initial value of GDP per worker tends to be associated with a higher subsequent growth rate in GDP per worker.

THE LONG-RUN GROWTH PROCESS



- Stylized fact 5:**
 Over periods of more than 130 years, many countries in Western Europe and North America have had relatively constant annual rates of growth in GDP per capita in the range 1.5-2 per cent.

- Labour's income share turns out to be quite stable (see figures)



- and labour's share of income can be written AS:

$$w_t / y_t$$

where w_t is the average real wage per worker, and y_t is GDP per worker.

- This implies:
- Stylized fact 6:** During the long periods of relatively constant growth rates in GDP per worker in the typical Western economy, labour's share of GDP has stayed relatively constant, hence the average real wage of a worker has grown by approximately the same rate as GDP per worker.

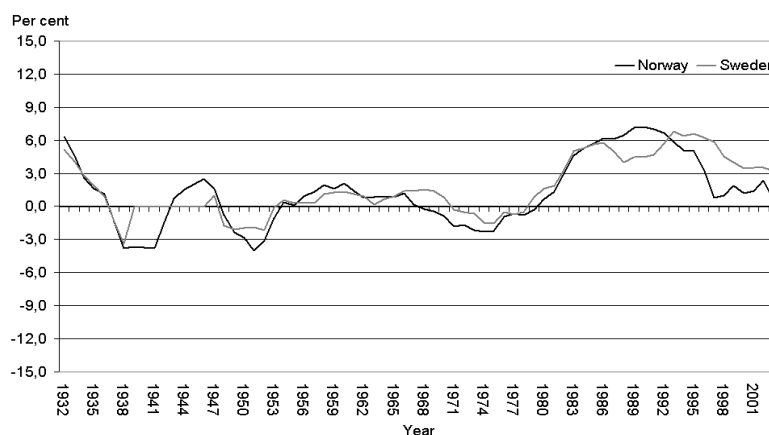
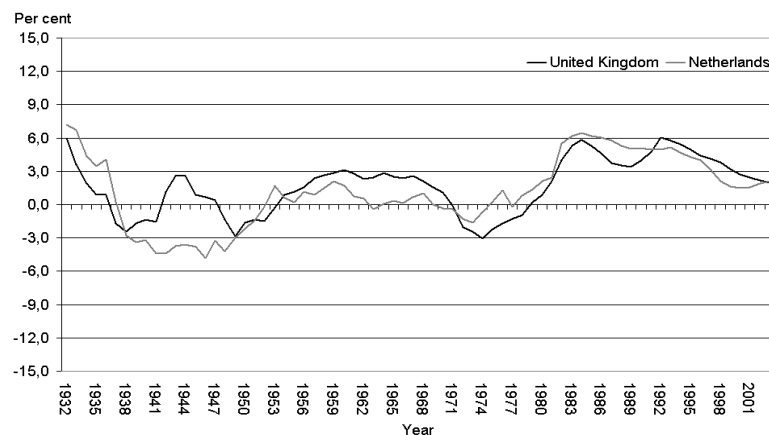
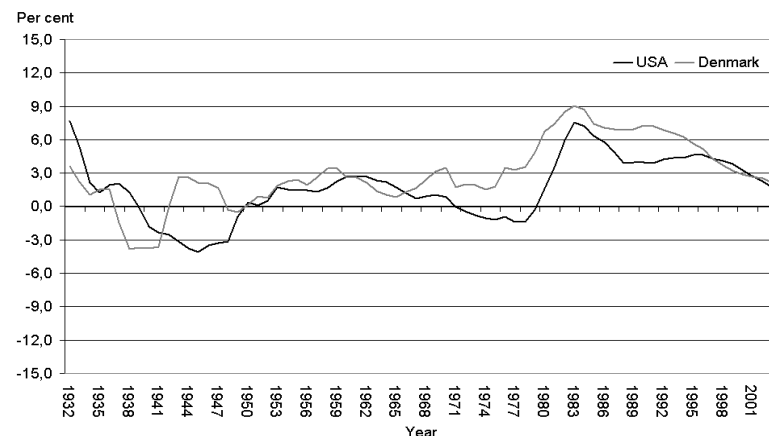
- The relative constancy of the factor income shares **and** the observation that real interest rates do not have any trend upwards or downwards in the long run (see figures), so that rates of return on capital must be relatively stable,

- and the fact that capital's share of income can be written as:

$$r_t / (Y_t / K_t)$$

where r_t is the rate of return on capital,

- implies:



- **Stylized fact 7:** During the long periods of relatively constant growth rates in GDP per worker in the typical Western economy, capital's share and the rate of return on capital have shown no trend, therefore the capital-output ratio K/Y has been relatively constant, and the capital intensity K/L has grown by approximately the same rate as GDP per worker.

BALANCED GROWTH

- summarizes stylized facts 5-7 and is an important concept for evaluating the growth models in the subsequent chapters.
- A growth process follows balanced growth if:
 1. GDP per worker, consumption per worker, the real wage and the capital intensity all grow at one and the same constant rate, g .
 2. The labour force (population) grows at a constant rate, n .
 3. GDP, consumption and capital grow at the common rate, $g+n$.
 4. The capital-output ratio and the rate of return on capital are constant.